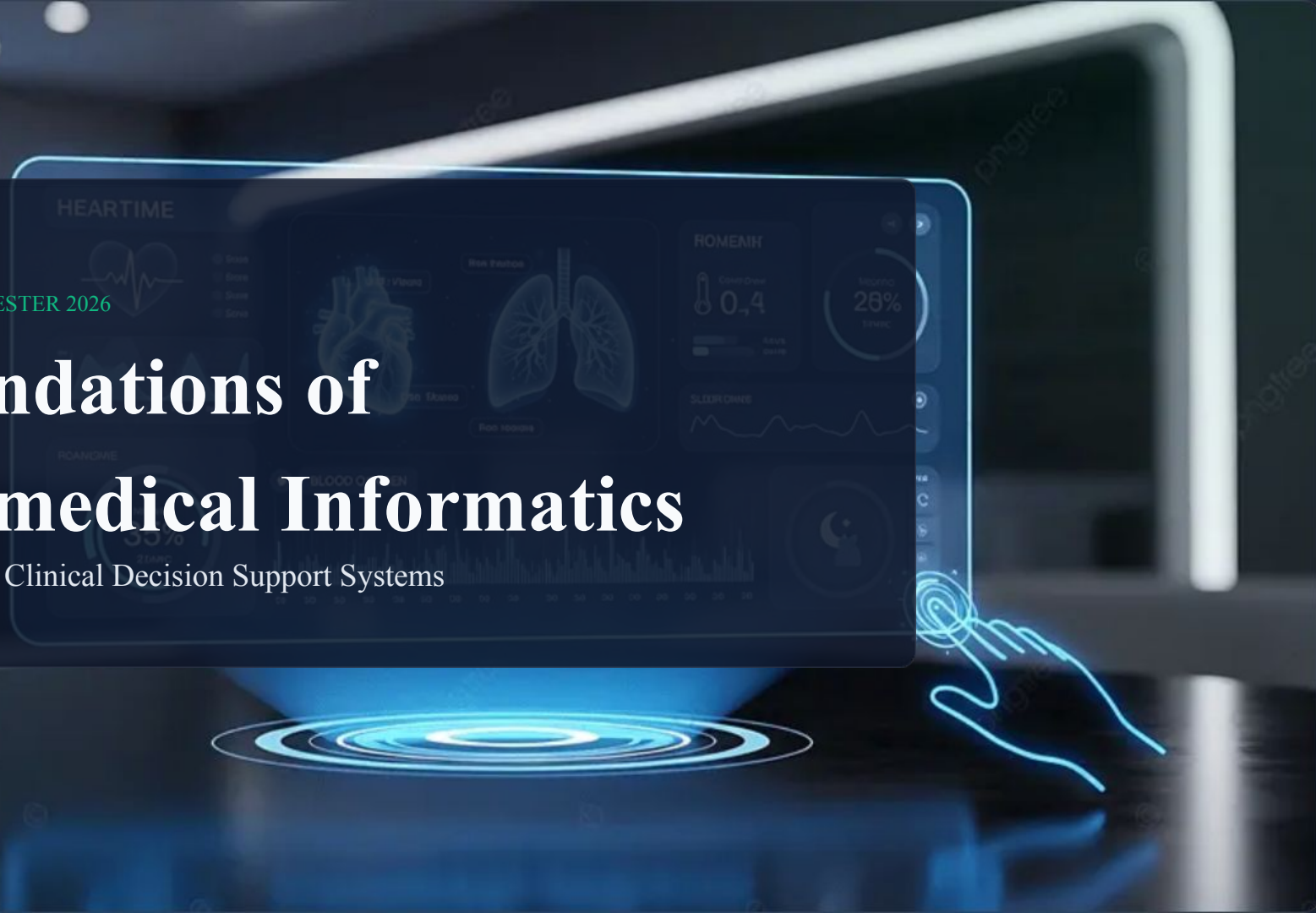


WINTER SEMESTER 2026

Foundations of Biomedical Informatics

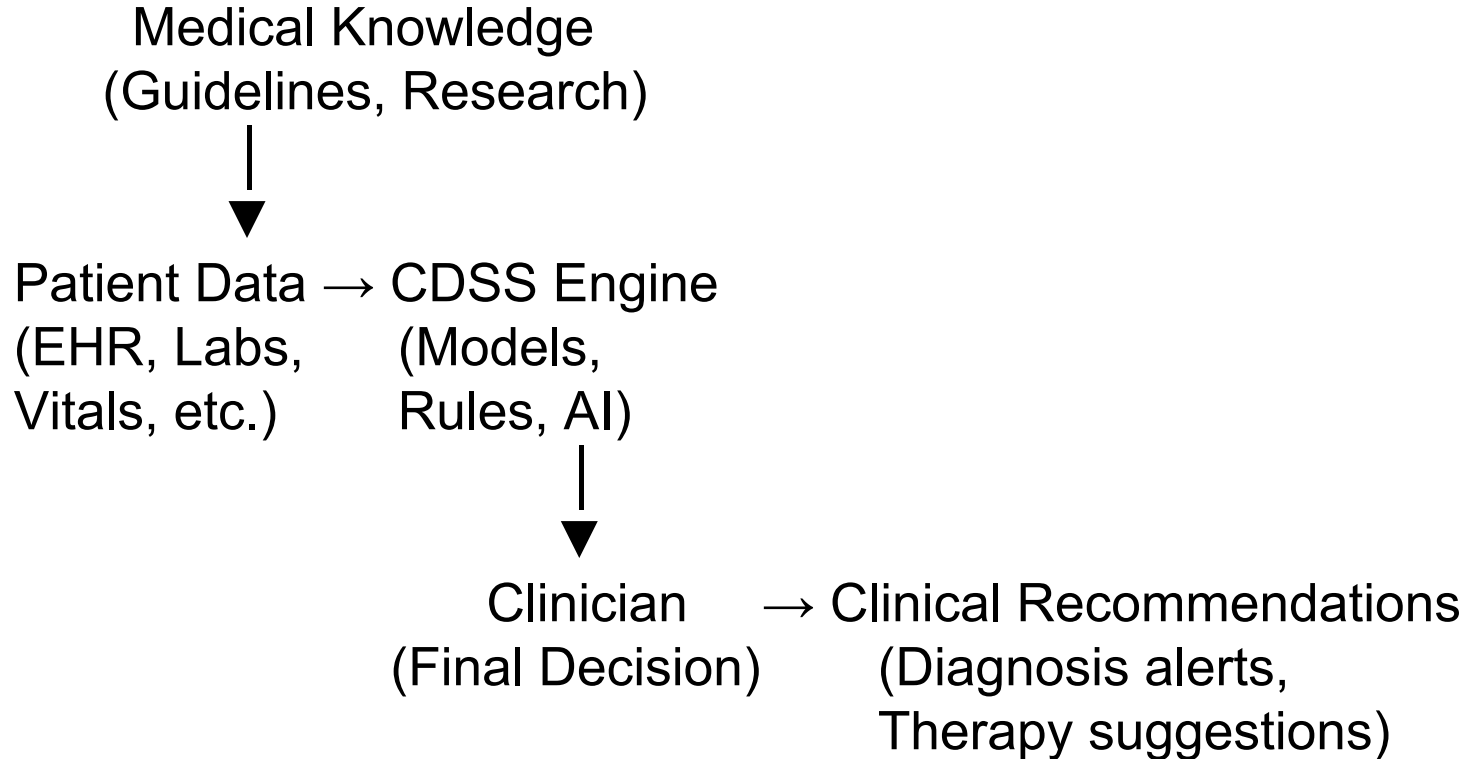
Lecture 10: Clinical Decision Support Systems



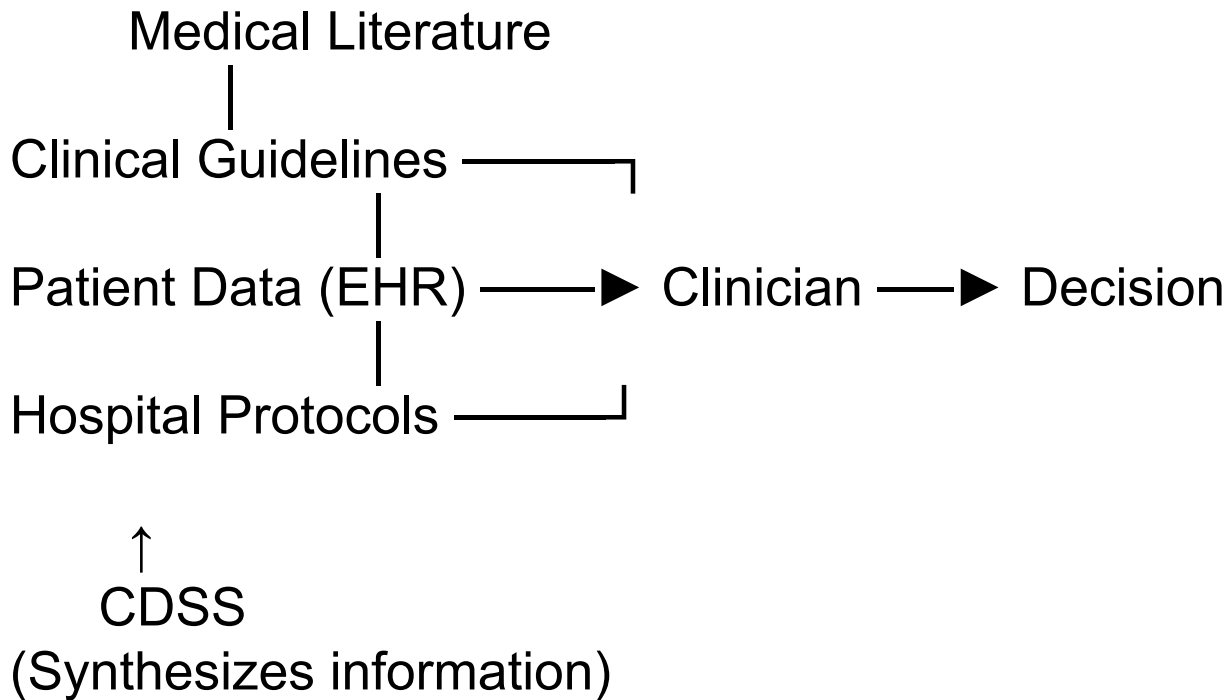
Clinical Decision Support Systems

- **Clinical Decision Support Systems (CDSS)** are computer programs designed to assist healthcare professionals in clinical decision making.
- They integrate **patient data, medical knowledge, and algorithms** to provide recommendations.
- Aim to improve:
 - Diagnostic accuracy
 - Treatment selection
 - Patient safety
 - CDSS supports clinicians but **does not replace physician judgement**.
- Increasingly integrated with **Electronic Health Records (EHRs)** and hospital information systems.

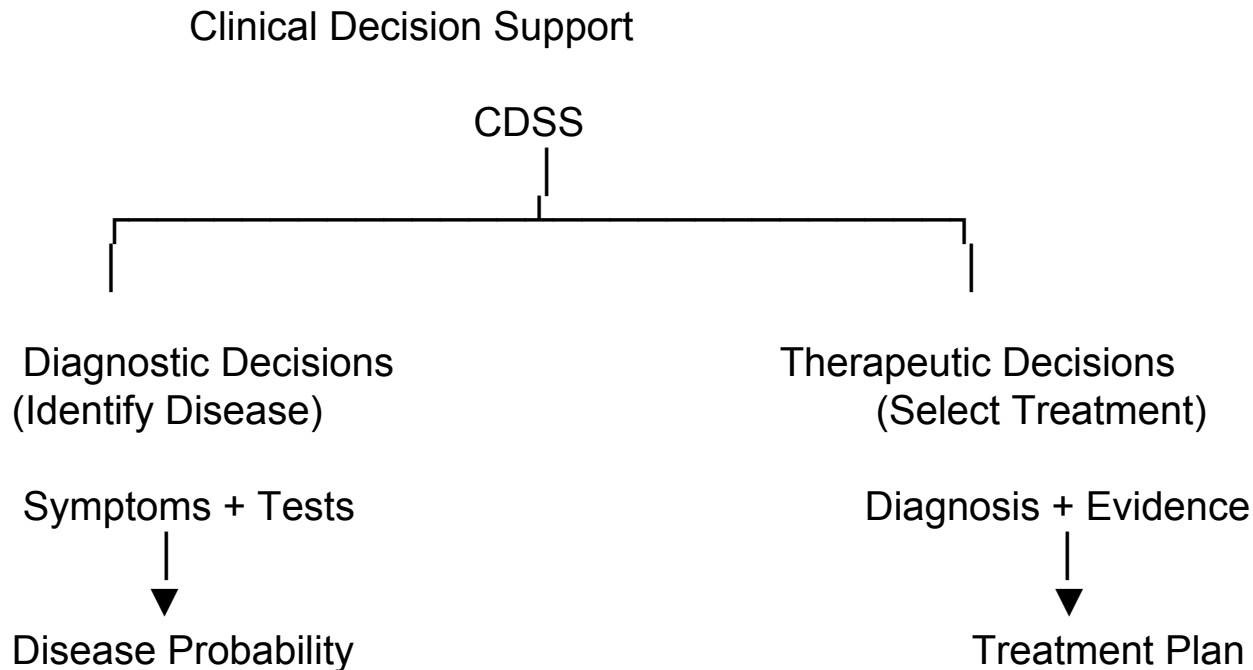
Clinical Decision Support Systems



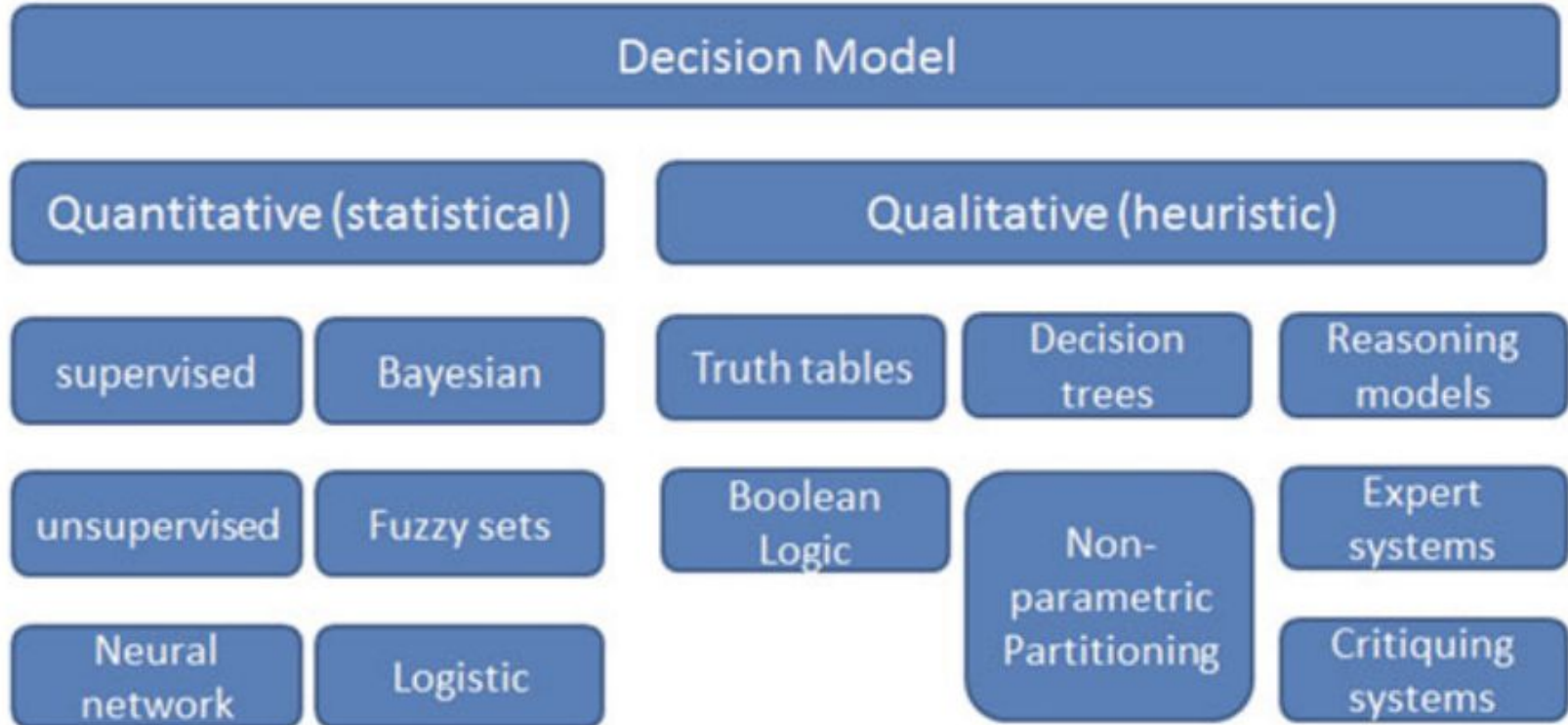
Clinical Decision Support Systems



Clinical Decision Support Systems



Clinical Decision Support Systems



Key Challenges in Building CDSS

1. **Complexity of Medicine:** Clinical data is **uncertain, incomplete, and probabilistic**
2. **Knowledge Acquisition:** Converting medical expertise into **computable models**
3. **System Architecture:** Designing systems for:
 - Diagnosis
 - therapy planning
 - patient monitoring
4. **Human-Computer Interaction:** Clinicians must trust and easily use the system.
5. **Integration:** CDSS must integrate with:
 - hospital information systems
 - EHR platforms
 - clinical workflows

Summary: Challenges in Building CDSS

1. Knowledge encoding
2. Uncertainty in data
3. Workflow integration
4. Usability for clinicians

First Generation CDSS: MYCIN

MYCIN was programmed in Lisp and used judgmental rules with associated elements of **uncertainty**. It was designed to identify bacteria causing severe infections (bacteremia, meningitis), and to recommend antibiotics, with the dosage adjusted for the patient's body weight. Edward Shortliffe, both a physician and computer scientist was confronted with problems associated with diagnosing a certain class of brain infections that was an appropriate area for expert system research and an area of particular importance, because the first 24 h are most critical for the patients. In the slide we see the idea of the separation of static knowledge (the rules and facts) and dynamic knowledge (the entries made by the human user and deductions made by the system). This is the principle of rule-based systems (Shortliffe and Buchanan [1984](#)).

Fate of MYCIN

MYCIN was not a success in the clinical practice; however, it was a pioneering work for practically each following system; for example ONCOCIN evolved from this work and assisted physicians in managing complex drug regimens for treating cancer patients. It has been built on the results of the MYCIN experiments while gaining experience with regular clinical use of an advice system for use by physicians. The work has also been influenced by data regarding features that may be mandatory if decision support tools are to be accepted by clinicians. Clinical oncology was selected due to the fact that this medical domain meets many of the criteria that have been identified for building an effective consultation tool using AI techniques (Shortliffe [1986](#)). Up to date, the main architecture of a DSS is the same as that developed in the 1970s.